Multimodal Approach for Heart Failure Risk Stratification

***Abstract*— Heart Failure (HF) is a highly prevalent cardiovascular disease affecting many people worldwide, and it represents a leading cause of hospitalization and mortality. The aim of this work is to present a multimodal system designed to support physicians in the management of patients already affected by HF, with a specific focus on risk stratification for the occurrence of adverse events. The system integrates heterogeneous data from Electronic Health Records (EHR), Electrocardiogram (ECG) signals, and Echocardiographic examinations to estimate the individual risk of adverse outcomes. This multimodal approach facilitates timely and personalized interventions, that improve patient outcomes and quality of life while reducing the burden on healthcare systems.**

Keywords — Heart failure, multiclass classification, risk stratification, multimodality, Electronic Health Records, Echocardiograms, Echography.

I. INTRODUCTION

Heart Failure (HF) is a very common cardiovascular condition and it is one of the main causes of death and hospitalization in industrialized countries. Its prevalence increases with age and exceeds 10% in people older than 70. HF is a complex clinical syndrome characterized by symptoms such as shortness of breath and fatigue, often associated with fluid retention, which significantly reduce patients' quality of life [1], [2].

This condition is caused by structural and functional changes in the heart that reduce its ability to pump blood effectively or increase pressure inside the heart. In response to an initial injury, the heart activates compensatory mechanisms that, over time, become harmful and lead to changes in ventricular size and function. These changes are also reflected in alterations of the heart's electrical activity [1], [2], [3], [4].

Patients with HF have a high risk of adverse events, including death, acute coronary syndrome, myocardial infarction, and the need for invasive interventions such as mechanical ventilation or emergency dialysis. Acute worsening episodes are particularly serious and are one of the main reasons for emergency department visits and hospital admissions.

Physicians often misclassify patient risk, discharging high-risk individuals too early while unnecessarily hospitalizing patients who are actually at low risk [5]. Rehospitalization rates are estimated to be between 30% and 60% within 3–6 months after discharge [6].

In this context, artificial intelligence (AI) is becoming increasingly important for risk assessment and for the management of patients with HF. The analysis of clinical and diagnostic data allows early identification of patients at higher risk of clinical deterioration and rehospitalization, supporting better decisions about hospitalization, discharge, and follow-up intensity [7].

The use of advanced predictive models can reduce unnecessary hospital admissions and allow timely intervention in high-risk patients, improving clinical outcomes and making the management of HF more efficient and economically sustainable.

II. DATA ACQUISITION AND REPRESENTATION

To support risk stratification of acute events in patients with established HF, this study relies on data collected from patients already affected by the disease at the time of observation. The dataset includes individuals hospitalized with new or worsening HF, for whom clinical, ECG, and imaging data were collected at a single baseline time point.

The target class for supervised learning is defined according to the patient's clinical status assessed six months after the baseline visit, reflecting the occurrence and severity of adverse outcomes. Rather than relying on continuous longitudinal monitoring, this approach associates baseline patient characteristics with outcome categories determined at a fixed follow-up time, enabling the development of risk stratification models that estimate short-term prognosis.

A. Electronic Health Records and Clinical Parameters (EHR)

EHRs provide a comprehensive representation of patients with established HF, capturing clinical, laboratory, and therapeutic factors that contribute to the risk of acute deterioration. Structured EHR data typically include demographics, comorbidities, vital signs, medication use, severity scores, laboratory biomarkers reflecting renal function, inflammation, electrolyte imbalance, and end-organ perfusion [8], [9]. EHR data complement signal and imaging-based modalities by capturing systemic and non-cardiac contributors to acute events, such as renal dysfunction, infection, and treatment intensity, which are not directly observable from ECG or Echocardiography alone [9].

However, EHR data are often limited by frequent missing values and inconsistent reporting formats, which necessitate robust data cleaning and imputation strategies.

B. Electrocardiographic Signals (ECG)

In patients with HF, the 12-lead ECG provides a widely available way to stratify risk and identify potential acute events before they occur [10]. ECG signals capture dynamic alterations in cardiac electrical activity associated with myocardial ischemia, conduction abnormalities, arrhythmias, and ventricular remodeling, all of which are known triggers or markers of acute HF decompensation [11], [12]. Specific ECG features such as QRS prolongation, QTc interval abnormalities, ST-segment deviations, T-wave inversion, and rhythm disturbances have been associated

with increased short-term and long-term mortality, as well as a higher risk of hospitalization in HF populations [10], [12].

ECGs represent an ideal tool given their non-invasive nature and widespread availability [11], [12]. Yet, they often lack specificity: different pathological processes can result in similar electrical changes, making it difficult to pinpoint the cause of a patient's decline [12]. Thus, while providing indirect information about cardiac structure and hemodynamics, the ECG is most effective when used in conjunction with imaging and other clinical data [9].

C. *Echocardiography Imaging*

Echocardiography is essential for stratifying risk in HF patients, as it provides real-time, non-invasive assessment of cardiac structure, function and hemodynamics [2], [13], [14]. Transthoracic Echocardiography (TTE) allows quantification of ventricular volumes, ejection fraction, wall thickness, and chamber remodeling, which are key determinants of vulnerability to acute decompensation [14].

Echocardiography is uniquely suited to identify hemodynamic abnormalities driving clinical deterioration, including elevated filling pressures, valvular dysfunction, pulmonary hypertension, and right ventricular failure [14]. Being portable and available at the bedside makes it useful especially for monitoring patients at high risk of decompensation [2]. However, Echocardiography data are affected by operator dependency and variability in image quality [15], which limit reproducibility and standardization across large datasets. Moreover, TTE acquisition can be technically challenging in acutely unwell patients due to severe dyspnea, tachycardia, or limited patient cooperation, often resulting in incomplete or focused examinations [14], [15]. Despite these limitations, Echocardiography remains a highly informative data modality for HF risk stratification.

III. MULTIMODALITY INTEGRATION

Given the heterogeneous nature of the data considered in this framework to assess heart failure risk stratification, a multimodal approach is required, employing dedicated learning architectures tailored to the specific characteristics of each modality.

A. *EHR-Based Architecture*

As illustrated in the EHR-based architecture (*Fig. 1*), raw clinical data undergo a series of transformations to generate the quantitative features required by the model. First, Min-Max normalization is applied, followed by recursive feature elimination (RFE) based on five-fold cross validation (CV) to reduce dimensionality. To ensure low variance and prevent overfitting, the framework utilizes extreme gradient boosting (XGBoost), an algorithm selected for its superior performance on tabular datasets [9]. The XGBoost decision tree algorithm uses second-order Taylor series to approximate the value of the loss function and it handles multi-class classification through an iterative process [9]. At each iteration, the model constructs three separate decision trees, one for each target class, to capture class-specific patterns. These outputs are then integrated through a symmetric multiple logistic transform to obtain a precise estimation of the final risk class probabilities [16].

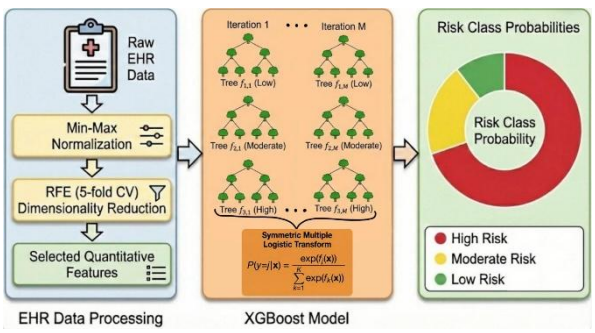

Figure 1: EHR-based architecture

B. *ECG-Based Architecture*

The proposed model (*Fig.2*) selects the first lead of the standard ECG trace sampled in segments lasting 30 seconds. The signal was initially subjected to a pre-processing phase; in particular, a Butterworth bandpass filter, with a cut-off frequency between 0.05 Hz and 45 Hz, was applied to remove high-frequency noise and correct baseline drift (baseline-wander) [17]. The architecture of the Deep Learning (DL) model used for feature extraction from the ECG is a One-Dimensional Residual Neural Network (1D ResNet) integrated with Squeeze-and-Excitation (SE) blocks. The 1D ResNet starts with a convolutional layer followed by four residual blocks, which are used to extract increasingly complex information from the signal. The internal structure of each block consists of two successive repetitions of the following sequence: 1D convolution, batch normalization, and Rectified Linear Unit (ReLU) activation; this sequence is followed by a SE module, which is applied at the end of each of the four residual blocks [17]. The 1D convolution layer is able to capture essential local morphological patterns (such as the morphology of P waves or the QRS complex), batch normalisation intervenes to stabilise the distribution of internal inputs, thus accelerating the convergence of the network, while ReLU introduces the necessary non-linearity to learn complex relationships that cannot be represented linearly. The SE module at the end, applies a mechanism that dynamically recalibrates the weights of the feature map channels emphasising the most informative characteristics and attenuating the irrelevant ones [17].

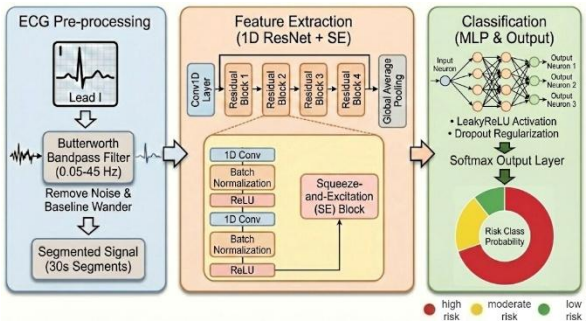

Figure 2: ECG-based architecture

The features extracted from ResNet are aggregated using Global Average Pooling and transmitted to a classification stage consisting of a Multi-Layer Perceptron (MLP). The MLP comprises fully connected layers that use the LeakyReLU activation function, a variant of ReLU that allows neurons receiving negative values to continue learning [17]. To improve the generalisation of the model, Dropout was applied, which randomly deactivates some of the neurons during training, preventing the network from overfitting. The output layer consists of three neurons associated with the Softmax activation function; the latter transforms the raw output scores into a normalised probability distribution [17].

C. *Echocardiography-Based Architecture*

The proposed framework (*Fig.3*) directly processes raw Echocardiographic videos, allowing the model to learn clinically relevant information from image data without relying on manually engineered features. All videos are temporally normalised to a fixed frame rate of 30 frames per second and spatially resized according to the specific Echocardiographic view. For each video, a fixed number of frames is selected to ensure consistent input dimensions across samples while preserving the temporal dynamics of the cardiac cycle.

The architecture utilizes a three-dimensional convolutional neural network (3D CNN) to aggregate spatial and temporal data throughout the network, facilitating the extraction of dynamic patterns related to cardiac anatomy and function [15].

Each unit follows a seven-layer sequence: two successive blocks of 3D convolution, batch normalization, and ReLu, functioning as previously described, concluding with a 3D Max Pooling operation. Specifically, the 3D convolution employs 3x3x3 kernels to scan the input volume and extract spatiotemporal features, capturing both anatomical details and motion across consecutive frames [15]. To highlight the most robust features while reducing dimensionality, the 3D Max Pooling layer down-samples the input by retaining only the maximum value within each 3x3x3 window [15].

A constrained low-parameter design is adopted to mitigate overfitting; since high-capacity models tend to memorize specific training examples, limiting the number of trainable weights ensures the learning of robust, generalizable patterns instead [15].

Following the convolutional units, the resulting 3D feature maps are flattened into a vector and fed directly

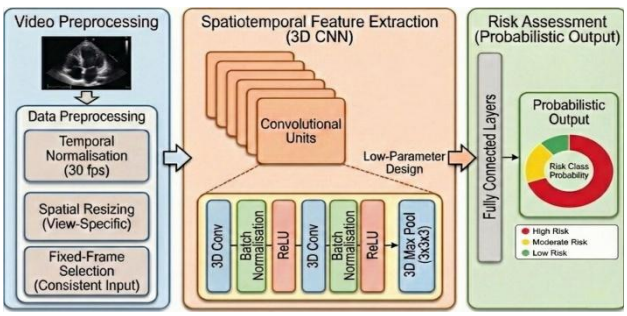

Figure 3: Echocardiography-based architecture

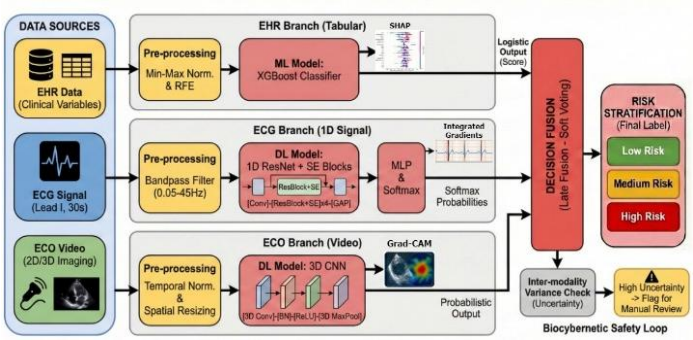

Figure 4: Multimodal pipeline for heart failure risk stratification

into a final fully connected layer. This layer takes the extracted high-level features as input and performs a weighted combination to return a single output value, representing the final probability scores [15].

D. *Decision-Level Fusion Architecture*

To integrate the results derived from these models, a Decision Level Fusion (Late Fusion) approach was chosen (*Fig.4*). This technique allows computational independence to be maintained and each branch to be optimised separately. The final classification is achieved through a Soft Voting mechanism, calculated using the formula (1), that averages the predict class probabilities. With the confidence level of each modality, this method ensure that highly confident predictions have a greater impact on the final decision, thereby improving the robustness of the risk assessment.

$$\hat{y} = \arg max_j \left(\frac{1}{N} \sum_{i=1}^{N} P_i(y = j) \right) \qquad (1)$$

where:

- $\hat{y}$ is the final predicted label;
- N is the total number of input modalities (in our case, N=3);
- p_{ij} is the softmax probability output by model i for class j.

E. *Evaluation Metrics*

The choice of evaluation metrics is driven by the clinical gravity of the classification task. While accuracy provides a general overview, class-specific precision, recall, and F1-score (2) are necessary to capture performance across different risk levels. F1-score is prioritized as the most representative metric, as it provides a harmonic balance between clinical safety (recall) and resource optimization (precision). While maximizing high-risk recall is fundamental to prevent dangerous clinical decisions and poor patient outcomes, the F1-score ensures that the system remains viable for hospital management: it prevents the model from generating excessive false alarms, which would otherwise lead to unnecessary hospitalizations and a strained healthcare system [18].

$$\boldsymbol{F1 - score} = \frac{2 \times precision \times recall}{precision + recall} =$$

$$= \frac{2 \times True\ Positives}{2 \times True\ Positives + False\ Positives + False\ Negatives} \qquad (2)$$

IV. TASK SELECTION

The proposed framework (*Fig.4*) addresses heart failure management through a risk stratification task rather than disease diagnosis. It formulates a multi-class classification problem leveraging a hybrid approach of Machine Learning (ML) and DL to estimate the likelihood of adverse events in patients with established HF. By stratifying patients into low, medium, and high-risk categories, the model focuses on assessing patient stability and capturing latent vulnerabilities through multimodal integration. In this context, the system functions as a clinical decision support system by providing structured and objective risk valuations to support clinician judgement, rather than replacing it.

V. BIOCYBERNETICS LOOP

The biocybernetics loop (*Fig.5*) integrates AI-based risk stratification into the clinical workflow, enabling continuous interaction between data acquisition and clinical decision-making for patients with HF.

To support transparency and clinical interpretability, some explainability techniques are integrated: for structured EHR data, Shapley Additive exPlanations (SHAP) are used to rank clinical variables according to their contribution to the predicted risk class, allowing clinicians to immediately identify the most relevant physiological parameters [9]; for ECG signals, Integrated Gradients are applied to generate temporal saliency maps, identifying the specific waveform segments that contribute most to risk prediction [19]; for echocardiographic data, Gradient-weighted Class Activation Mapping (Grad-CAM) creates heatmaps to highlight anatomical structures influencing the classification [4], [13]. These explainability mechanisms allow clinicians to contextualize model outputs within established pathophysiological knowledge.

In addition to explainability, the framework incorporates uncertainty awareness by monitoring agreement across modalities during the late fusion stage. Inter-modality variance, defined as the dispersion among risk probabilities predicted by the EHR-, ECG-, and echocardiography-based architectures, is used as a proxy for predictive uncertainty. To prioritize clinical safety, any marked discordance among modalities flags the case for expert review, preventing reliance on automated results when model confidence is low.

By combining explainability and uncertainty quantification within a closed feedback loop, the proposed biocybernetics framework supports adaptive clinical decisions. In particular, it enables timely intervention for patients identified as high risk, while reducing unnecessary hospitalizations in individuals likely to remain stable, thereby contributing to more efficient and safer management of HF.

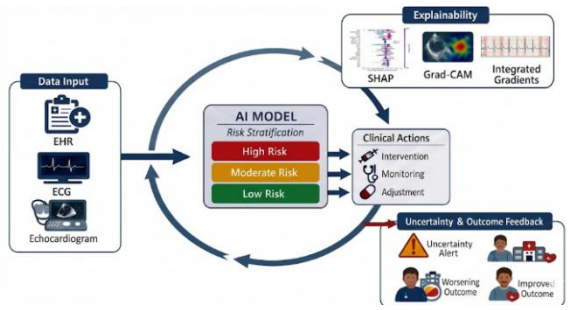

Figure 5: Biocybernetics Loop

REFERENCES

[1] T. A. McDonagh *et al.*, «2021 ESC Guidelines for the diagnosis and treatment of acute and chronic heart failure», *Eur. Heart J.*, vol. 42, fasc. 36, pp. 3599–3726, set. 2021, doi: 10.1093/eurheartj/ehab368.

[2] E. Di Cesare *et al.*, «Multimodality imaging in chronic heart failure», *Radiol. Med. (Torino)*, vol. 126, fasc. 2, pp. 231–242, feb. 2021, doi: 10.1007/s11547-020-01245-4.

[3] X.-M. Li *et al.*, «Electrocardiogram-based artificial intelligence for the diagnosis of heart failure: a systematic review and meta-analysis».

[4] C.-K. Lee *et al.*, «Multimodal deep learning models utilizing chest X-ray and electronic health record data for predictive screening of acute heart failure in emergency department», *Comput. Methods Programs Biomed.*, vol. 255, p. 108357, ott. 2024, doi: 10.1016/j.cmpb.2024.108357.

[5] S. P. Collins e P. S. Pang, «ACUTE Heart Failure Risk Stratification: A Step Closer to the Holy Grail?», *Circulation*, vol. 139, fasc. 9, pp. 1157–1161, feb. 2019, doi: 10.1161/CIRCULATIONAHA.118.038472.

[6] I. G. Stiell *et al.*, «Prospective and Explicit Clinical Validation of the Ottawa Heart Failure Risk Scale, With and Without Use of Quantitative NT -pro BNP», *Acad. Emerg. Med.*, vol. 24, fasc. 3, pp. 316–327, mar. 2017, doi: 10.1111/acem.13141.

[7] F. Yasmin *et al.*, «Artificial intelligence in the diagnosis and detection of heart failure: the past, present, and future», *Rev. Cardiovasc. Med.*, vol. 22, fasc. 4, p. 1095, dic. 2021, doi: 10.31083/j.rcm2204121.

[8] Z. Chen, T. Li, S. Guo, D. Zeng, e K. Wang, «Machine learning-based in-hospital mortality risk prediction tool for intensive care unit patients with heart failure», *Front. Cardiovasc. Med.*, vol. 10, p. 1119699, apr. 2023, doi: 10.3389/fcvm.2023.1119699.

[9] K. Wang *et al.*, «Interpretable prediction of 3-year all-cause mortality in patients with heart failure caused by coronary heart disease based on machine learning and SHAP», *Comput. Biol. Med.*, vol. 137, p. 104813, ott. 2021, doi: 10.1016/j.compbiomed.2021.104813.

[10] J. Václavík *et al.*, «ECG in patients with acute heart failure can predict in-hospital and long-term mortality», *Intern. Emerg. Med.*, vol. 9, fasc. 3, pp. 283–291, apr. 2014, doi: 10.1007/s11739-012-0862-1.

[11] A. H. Ribeiro *et al.*, «Automatic diagnosis of the 12-lead ECG using a deep neural network», *Nat. Commun.*, vol. 11, fasc. 1, p. 1760, apr. 2020, doi: 10.1038/s41467-020-15432-4.

[12] J. K. Zègre-Hemsey *et al.*, «Prehospital ECG with ST-depression and T-wave inversion are associated with new onset heart failure in individuals transported by ambulance for suspected acute coronary syndrome», *J. Electrocardiol.*, vol. 69, pp. 23–28, nov. 2021, doi: 10.1016/j.jelectrocard.2021.08.005.

[13] A. Valsaraj *et al.*, «Development and validation of echocardiography-based machine-learning models to predict mortality», *eBioMedicine*, vol. 90, p. 104479, apr. 2023, doi: 10.1016/j.ebiom.2023.104479.

[14] S. Fitzsimons e R. N. Doughty, «Role of transthoracic echocardiogram in acute heart failure», *Rev. Cardiovasc. Med.*, vol. 22, fasc. 3, p. 741, set. 2021, doi: 10.31083/j.rcm2203081.

[15] A. E. Ulloa Cerna *et al.*, «Deep-learning-assisted analysis of echocardiographic videos improves predictions of all-cause mortality», *Nat. Biomed. Eng.*, vol. 5, fasc. 6, pp. 546–554, feb. 2021, doi: 10.1038/s41551-020-00667-9.

[16] J. H. Friedman, «Greedy function approximation: A gradient boosting machine.», *Ann. Stat.*, vol. 29, fasc. 5, ott. 2001, doi: 10.1214/aos/1013203451.

[17] S. González *et al.*, «Multi-modal heart failure risk estimation based on short ECG and sampled long-term HRV», *Inf. Fusion*, vol. 107, p. 102337, lug. 2024, doi: 10.1016/j.inffus.2024.102337.

[18] D. B. Olawade *et al.*, «Smart emergency care: a narrative review of predictive machine learning models», *Ann. Transl. Med.*, vol. 13, fasc. 5, pp. 60–60, ott. 2025, doi: 10.21037/atm-25-83.

[19] L. Wei e Y. Li, «A multi-scale CNN-Transformer parallel network for 12-lead ECG signal classification», *Signal Image Video Process.*, vol. 19, fasc. 8, p. 611, ago. 2025, doi: 10.1007/s11760-025-04215-3.